

ADITYA PANDEY

B.Tech CSE, 2027 | Noida, Uttar Pradesh

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SUMMARY

Full-Stack Developer building AI-powered systems with MERN, Next.js, FastAPI, LangChain, and LangGraph, specializing in agentic workflows and RAG-based semantic retrieval. LeetCode rated 1835 (Top 6% globally) with 800+ DSA problems solved.

TECHNICAL SKILLS

Languages : Java, Python, JavaScript, TypeScript, SQL

Frontend : Next.js, React.js, Tailwind CSS, ShadCN UI, Redux Toolkit, RTK Query, HTML5, CSS3

Backend & Databases : Node.js, Express.js, FastAPI, REST APIs, MongoDB, PostgreSQL, Supabase, Mongoose, JWT

AI & ML : LangChain, LangGraph, RAG, NLP, Sentence-Transformers, Hugging Face, Groq

Developer Tools : Git, GitHub, Vercel, Render, Postman, MongoDB Atlas, Cloudinary, Figma, Clerk

CS Fundamentals : Data Structures & Algorithms, OOPs, Operating Systems, DBMS

PROJECTS

JobPilot.AI – Agentic Job Discovery Platform | 🧠🔍🔗 | *FastAPI, LangChain, LangGraph* **Aug 2026 – Sep 2026**

- Architected an **agentic job discovery engine** using **FastAPI, LangChain, and Groq LLMs** to parse conversational intent and route requests across **Adzuna and Greenhouse APIs**, ranking results via semantic relevance, resume fit, and real-time company trust signals (**Tavily**).
- Engineered a deterministic **pre-filtering layer** enforcing location and seniority bounds before LLM evaluation, cutting inference token consumption by **70%+**, then processed matches through a fault-tolerant **batch pipeline** with structured **Pydantic** outputs and rate-limit handling for **sub-6-second** search latency.
- Built a deterministic **canonical-key deduplication engine** eliminating **100% of exact overlaps** across job sources while retaining the highest-relevance match, paired with a **Manifest V3 Chrome Extension** that parses live ATS pages for real-time resume-fit analysis.

EmbedAI - PDF AI Chatbot | 🧠🔍🔗 | *Next.js, TypeScript, Supabase, pgvector, Clerk* **Mar 2026**

- Architected a **RAG pipeline** with parallelized batch embedding generation and **HNSW-indexed pgvector search**, reducing document processing time by **3x** and achieving **sub-100ms** semantic retrieval.
- Engineered a lightweight **JavaScript widget** served via **Next.js API routes** with dynamic CORS handling, enabling businesses to embed isolated AI chatbots through a single script tag.
- Designed a secure **3-tier RBAC system** using **Clerk Webhooks (Svix HMAC)** and **Supabase Row Level Security**, ensuring multi-tenant data isolation with **tiered API rate limiting (HTTP 429)**.

AI-Grievance Intelligence System | 🧠🔍 | *MERN, Python, FastAPI, Sentence-Transformers* **Jan 2026 – Feb 2026**

- Architected a hybrid microservices system using **FastAPI** for AI processing and **Node.js** for core logic, with **4-tier RBAC** and immutable audit logs for secure, asynchronous complaint processing.
- Engineered a semantic deduplication layer using **vector embeddings** to block spam and redundant reports, cutting duplicate database writes by **30%**, paired with a **Human-in-the-Loop** retraining pipeline that improves routing accuracy through admin corrections.

ScholarHub - LMS | 🧠🔍🔗 | *React, Node.js, Express.js, MongoDB, Redux Toolkit* **Aug 2025 – Dec 2025**

- Engineered a full-stack LMS with **React, Node.js, and RTK Query** caching, reducing redundant API calls by **35%** while integrating **Stripe** payments and **Cloudinary** for secure payments and video delivery.
- Built an AI-powered “Intelligent Tutor” using the **Gemini API** for auto-generated quizzes and notes, and optimized course discovery with **MongoDB Compound Indexing**, reducing query execution time by **40%**.

ACHIEVEMENTS & CODING

- Problem Solving:** Achieved a **Max Rating of 1835** on LeetCode (Top 6% globally), a **3-Star Rating** on CodeChef, and solved **800+ DSA problems** across multiple platforms.
- DevsOnLoop Hackathon (2025):** Secured **3rd Place** by building a **Maintenance Complaint Portal**, recognized for “Best Practical Implementation”.
- Synergia (2024):** Secured **1st Position** in a technical presentation event for communicating engineering concepts.

EDUCATION

Bundelkhand Institute of Engineering and Technology

Bachelor of Technology in Computer Science & Engineering; CGPA: 7.97

Jhansi, Uttar Pradesh

2023 – 2027